Ella Majkic

Project Portfolio: emajkic.github.io
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Education

The University of British Columbia Sept 2023 - Present

BASc in Engineering Physics — Minor in Honours Mathematics

- **Awards:** Trek Excellence Scholarship; Eric P. Newell Engineering Award; Dean's Scholar **GPA: 90%**

Skill Summary

Software: C/C++, Python, Java, Git, MATLAB, Simulink, ROS, OpenCV, TensorFlow, STM32CubeIDE

Tools: Altium, FPGA/VHDL, 8051 Assembly, Linux, Terminal, COMSOL, CMake, OnShape, LaTeX, GDB

Experience

LUNR Aerospace Corp. May 2026 - Present

Engineering Intern, Rocket Electronics and Controls

- Developing embedded thrust vectoring control (TVC) system for directional control of liquid engine rocket
- Derived quaternion-based rocket body attitude estimation through simulation and numerical analysis in Python
- Built dynamic Simscape models from SolidWorks assemblies, validating gimbal positions to 0.01 mm accuracy
- Validated TVC PID loop convergence in Python (JupyterLab) and tuned full control scheme in Simulink
- Set up LabVIEW DAQ system for engine hotfire tests, controlling thermocouples, transducers, and valves

TRIUMF Particle Accelerator Jan 2025 - May 2025

Research Intern, PIONEER Experiment

- Developed and tested Purity Monitor Assembly (PUMA) calibration device for rare pion decay experiment
- Simulated electron drift dynamics using COMSOL and C++, validating physics models against collected data
- Drafted and built a vacuum system for PUMA calibration tests, achieving stable 1e-6 bar vacuum pressure
- Led first successful PUMA data collection tests, performing statistical analysis in Python and MATLAB
- Designed and soldered RS-232-based data acquisition for vacuum gauges and detector readout in LabVIEW

UBC Rocket Jan 2026 - Present

Hardware Engineer, Avionics Hardware Subteam

- Developed and tested hardware, firmware, and PCBs for single-stage 30km altitude competition rocket
- Wrote embedded firmware in C for a custom ignition PCB using an STM32 (ARM Cortex-M0+) MCU, allowing for communication with flight controller and precise logic for critical in-flight parachute deployment
- Developed PID control loop for ground station antennae to track rocket and receive in-flight telemetry (live video stream, GPS packets, velocity/acceleration data) based on RSSI values from RFD900x radio module

Projects [Portfolio]

Simulated Detective Agent

Deep Learning, TensorFlow, ROS, Computer Vision, CNN, Linux

- Trained and integrated multiple robust machine learning models from scratch to enable a ROS robot agent to autonomously solve a detective-style task in Gazebo simulation, achieving top score in course
- Designed and trained a convolutional neural network on a custom, augmented 1,000+ image dataset, achieving 99.1% validation accuracy on alphanumeric character recognition
- Used YOLOv8, OpenCV, and homography for dynamic clueboard and NPC detection within environment

Autonomous Pet Rescue Robot

C++, Electronics Design, Rapid Prototyping, PCB Assembly

- Developed, prototyped, and built a fully autonomous robot with a team capable of line following through a multi-terrain course while identifying and retrieving pet stuffies
- Designed and soldered electrical systems for motor control, microcontroller integration, and sensing
- Wrote C++ libraries to interface with 2D LiDAR sensors, H-bridges, servos, and inverse-kinematics claw